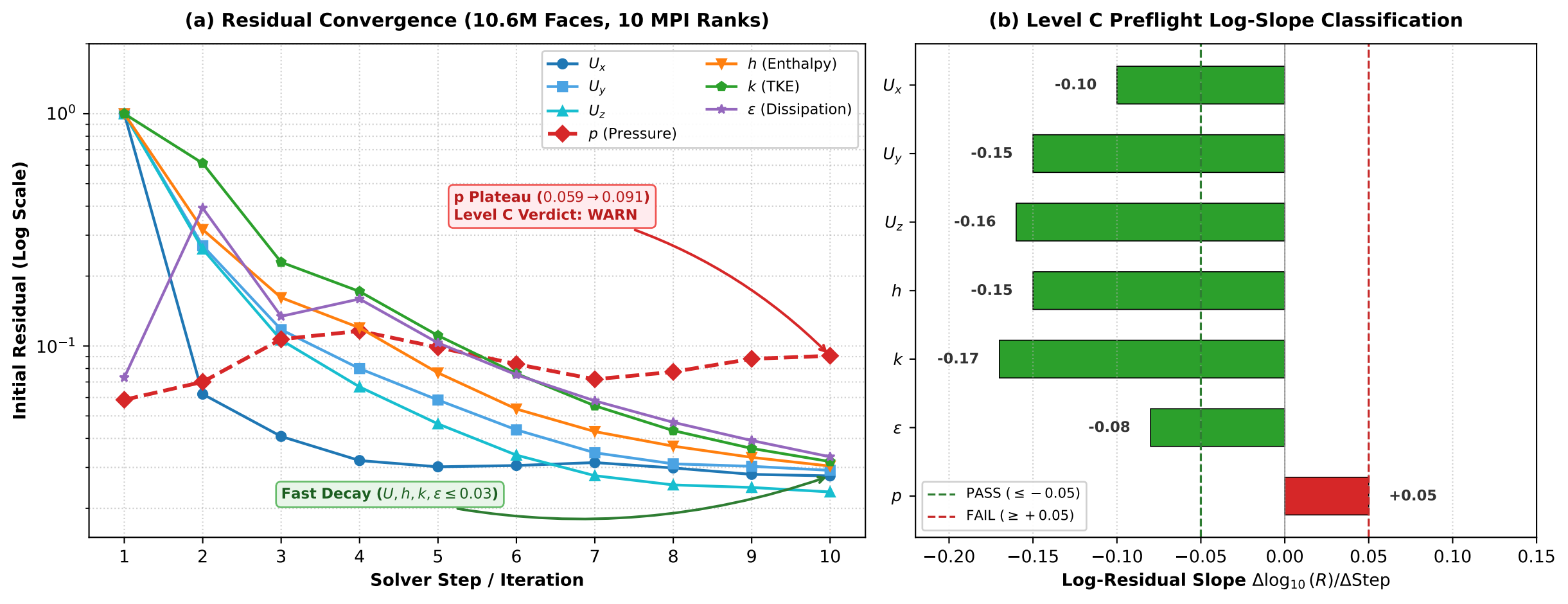
(a) Residual Convergence (10.6M Faces, 10 MPI Ranks)
Initial Residual (Log Scale)
10^0
10^{-1}
Solver Step / Iteration
1
2
3
4
5
6
7
8
9
10
U_x
U_y
U_z
p (Pressure)
h (Enthalpy)
k (TKE)
ε (Dissipation)
p Plateau (0.059 → 0.091)
Level C Verdict: WARN
Fast Decay ($U, h, k, \varepsilon \leq 0.03$)
(b) Level C Preflight Log-Slope Classification
U_x
U_y
U_z
h
k
ε
p
-0.10
-0.15
-0.16
-0.15
-0.17
-0.08
+0.05
PASS (≤ -0.05)
FAIL ($\geq +0.05$)
−0.20
−0.15
−0.10
−0.05
0.00
0.05
0.10
0.15
Log-Residual Slope $\Delta \log_{10}(R)/\Delta$Step